# AGV2 Kayaba — Function Summary

Differential-drive magnetic-tape AGV with front and rear MGS1600 sensors, RFID route tags, trolley-hitch pusher, lidar safety fields, keyboard / remote control and a Flask web HMI. Read from `profiles/agv2_kim.json` and the controller code (commit 85d18b0) · 15 Oct 2026. All RFID tag values are placeholders until replaced with the real scanned hex.

## 1. Operating modes

| Mode | Entered by | Behaviour |
|---|---|---|
| ARMED | Startup, RESET, arriving at homepost or B0 | Stopped, motors released, waiting for a start command. |
| RUNNING (AUTO forward) | Branch button DI4–7 / keys 7 8 9 0, CALL START (DI8), web START, or end of a reverse | Forward PID tape-following on the front MGS1600. |
| REVERSING | Tag 0001 (branch 2: tag 0002), tag 0003 (not branch 2, left track), web START REVERSE, or key R | Backward PID tape-following on the rear MGS1600. |
| MANUAL | Mode selector DI2 ON, key Q, or web bypass | Driving from the web jog pad, the laptop keyboard or the USB remote (WASD / arrows). |
| ROTATING | Key T / Y or web ROTATE 180° (from MANUAL or ARMED) | 180° spin-turn on the tape, then braked back in MANUAL / ARMED (section 11). |
| EMERGENCY | E-stop DI3, key E, or web E-STOP | All motion stops, motors released (AGV can be pushed). RESET (DI1 or ENTER) to recover. |

## 2. Ways to start and stop

- **Branch buttons DI4–7 or keys 7 / 8 / 9 / 0** — in ARMED: select TITIK 1–4 **and** start AUTO (normal run). In other modes they only select; ignored while driving.
- **CALL START (DI8)** — only at homepost (DO11 ON). That run stops at tag B0; continue with a branch button or web START.
- **No physical START button** — DI0 is the pusher button (`DI_START: -1`).
- **RESET** — DI1, the ENTER key / OK on the remote, or web RESET (bypass mode). Leaves EMERGENCY; aborts AUTO / REVERSE back to ARMED.
- **Web MANUAL page** — bypass mode gives START / RESET / MODE / E-STOP buttons (no password); START REVERSE starts REVERSE from ARMED.

# 3. Speeds

| Mode / setting | Value | Notes |
|---|---|---|
| APPROACH (tag 0012, key 1) | 0.1 m/s | APPROACH PID gains, eases down at 0.2 m/s² |
| SLOW (tag 0014, key 2) | 0.2 m/s | SLOW PID gains |
| EXTRA_SLOW (after a reverse) | 0.15 m/s | SLOW PID gains · set by the hitch stop (00CC / marker) when driving forward again; `AUTO_TARGET_EXTRA_SLOW_SPEED` on PARAMS |
| SLOW in branch (tags 00AA / 00AB / 00AC) | 0.2 m/s | **Own SLOW BRANCH PID gains** (`KP/TD/N/V_RED_COEF_SLOW_BRANCH`, start = SLOW values, own auto-tune option) until the next speed tag or resume |
| MEDIUM (tag 0015, key 3) | 0.4 m/s | **Own MEDIUM PID gains** (start = SLOW values: KP 2.2, TD 0.14, N 12, V-Red 2) |
| HIGH (tag 0016, key 4) | 0.6 m/s | HIGH PID gains (KP 1.4, TD 0.12, N 20, V_RED_COEF 4) |
| MAX (tag 0018, key 5) | 1.0 m/s | **Own MAX PID gains** (KP_MAX 1.0, TD_MAX 0.15, N_MAX 20, V_RED_COEF_MAX 4) · motors top out at ~0.94–0.98 m/s |
| Reverse | 0.15 m/s | 0.1 m/s after tag 00CA · 30 s safety timeout (paused while holding) |
| AUTO accel / decel | 0.2 / 0.2 m/s² | ACCEL_RATE / DECEL_RATE |
| Manual high / slow | 0.4 / 0.2 m/s | Slow speed used for turning |
| Manual accel / decel | 0.8 / 0.8 m/s² | Smooth start and stop; direction change passes through zero |
| Lidar outer / middle | 60 % / 30 % of the current speed | e.g. MAX 1.0 → 0.6 / 0.3 m/s · eases down at 0.2 m/s² (lidar is OFF by default) |

- Keys 1–5 set the AUTO speed mode in ARMED (starting speed of the next run) or while RUNNING; the next speed tag overrides them.
- Forward steering direction is flipped (`ERROR_SIGN_FORWARD: 1`).
- **Accel / decel rates are real m/s²** (fixed 19 Oct 2026): the ramp now uses the real time between sensor frames (~10 Hz). Before, it stepped a fixed 10 ms per frame, so every ramp was ~10× slower than set (decel 0.5 took ~4.5 m from HIGH to SLOW). Braking distance = $(v_1^2 - v_2^2) / (2 \times \text{decel})$: HIGH→SLOW at 0.1 m/s² = 1.6 m, at 0.5 m/s² = 0.3 m. With a 2.1 t trolley keep DECEL_RATE ≈ 0.1.
- Tape loss or sensor (CAN) timeout: brake, resume automatically when the tape is seen again.
- Near top speed the steering difference is kept: if a wheel would need more than 5 V, both wheels are slowed by the excess (MAX 1.0 m/s asks ~3180 rpm, the motors reach ~3130).
- Speeds, ramps and PID gains can be tuned on the PARAMS page (engineer login). **APPLY saves them as the new defaults** (kept after restart); RESET TO PROFILE goes back to the profile values.

## 4. RFID mapping

| Tag | Sequence | Fires when | Action | Cooldown |
|---|---|---|---|---|
| 0012 | speed_zone_approach | RUNNING | Speed → APPROACH (0.1 m/s) | 0 s |
| 0014 | speed_zone_slow | RUNNING | Speed → SLOW (0.2 m/s) | 0 s |
| 0015 | speed_zone_medium | RUNNING | Speed → MEDIUM (0.4 m/s) | 0 s |
| 0016 | speed_zone_high | RUNNING | Speed → HIGH (0.6 m/s) | 0 s |
| 0018 | speed_zone_max | RUNNING | Speed → MAX (1.0 m/s) | 0 s |
| 000C | release_at_c | RUNNING, **not on branch 2** | Trolley release — section 7 | 5 s |
| 001C | release_at_1c_branch2 | RUNNING + **branch 2 only** | Trolley release — section 7 | 5 s |
| 00AA | branch_titik_1 | RUNNING + branch 1 | SLOW (branch gains), track left | 5 s |
| 00AB | branch_titik_2 | RUNNING + branch 2 | SLOW (branch gains), track right | 5 s |
| 00A2 | branch2_track_left_a2 | RUNNING + **branch 2 only** | Track left (speed unchanged) | 5 s |
| 00AC | branch_titik_3 | RUNNING + branch 3 | SLOW (branch gains), track left | 5 s |
| 00AD | branch_titik_4 | RUNNING + branch 4 | SLOW only (no track action yet) | 5 s |
| 00AE | fork_always_right | RUNNING (every run) | Track left | 5 s |
| 00BB | branch_merge_reset | RUNNING (every run) | Track right (branch is **kept**) | 5 s |
| 00AE | reverse_fork_ae | **REVERSING** | Reverse track left (rear sensor) — branch 2, 4 / no branch | 5 s |
| 00BB | reverse_merge_bb | **REVERSING** | Reverse track right (rear sensor) — branch 2, 4 / no branch | 5 s |
| 0001 | reverse_at_1 | RUNNING, **not branch 2**, skipped if ignore_rfid_1 set | Stop, wait 2 s, start REVERSE | 3 s |
| 0002 | reverse_at_2_branch2 | RUNNING + **branch 2 only**, skipped if ignore_rfid_2 set | Stop, wait 2 s, start REVERSE: starts right, **00AE → left**, 00BB → right | 3 s |
| 0003 | reverse_at_3_left | RUNNING, every branch **except 2**, skipped if ignore_rfid_3 set | Stop, wait 2 s, start REVERSE on the **LEFT** track only (branch rule and AE/BB ignored) | 3 s |
| 00BA | arm_reverse_marker_ba | REVERSING | Arm DI10 stop marker (marker ignored before BA) | 3 s |
| 00CA | reverse_approach_ca | REVERSING | Reverse speed → 0.1 m/s | 3 s |
| 00CC | stop_at_c1 | REVERSING (also DI10 marker after 00BA) | Hitch sequence — section 6 | 5 s |
| 00CE | clear_ignore_at_cr | Any mode | Clear ignore_rfid_1 / 2 / 3 (tags 0001 / 0002 / 0003 work again) | 3 s |
| 00CB | ignore_reverse_tags_cb | Any mode | Set ignore_rfid_1 / 2 / 3 — reverse tags 0001 / 0002 / 0003 ignored until 00CE | 3 s |
| 0000 | arrive_homepost | RUNNING, not right after leaving home | Stop, **clear branch**, exit to ARMED, DO11 ON | 5 s |
| 00B0 | stop_at_b0 | RUNNING, CALL START runs only | Stop, exit to ARMED | 5 s |

- **Branch selection** (TITIK 1–4) stays set for the whole run and is cleared **only at homepost** (tag 0000, or reading the homepost tag while parked).
- Cooldown only stops the **same** tag firing twice in a row; it never delays the first read. While a stop sequence is running (C, 1C, CC, 1) other tags are ignored.

# 5. IO mapping

DIO module 192.168.1.31 (Modbus TCP, 16 DI / 16 DO). All DIs are inverted in software (`DI_FLIPPED = 1`): “ON” below is the value the code sees and the IO page shows; an unwired input reads ON.

## Digital inputs

| DI | Function | Meaning / use |
|---|---|---|
| 0 | PUSHER BUTTON | Press: full UP / full DOWN by position; press while moving = stop. ARMED or MANUAL only |
| 1 | RESET | Rising edge = reset / leave emergency (same as ENTER key) |
| 2 | MODE selector | ON = MANUAL, OFF = AUTO (key Q overrides until the selector is moved) |
| 3 | EMERGENCY | ON = e-stop tripped |
| 4 | BRANCH 1 button | Select TITIK 1 (+ start AUTO in ARMED) |
| 5 | BRANCH 2 button | Select TITIK 2 (+ start AUTO in ARMED) |
| 6 | BRANCH 3 button | Select TITIK 3 (+ start AUTO in ARMED) |
| 7 | BRANCH 4 button | Select TITIK 4 (+ start AUTO in ARMED) |
| 8 | CALL START | Start AUTO only at homepost; run stops at B0 |
| 9 | TROLLEY LIMIT 1+2 (NC) | Both switches in parallel · OFF = trolley caught |
| 10 | REVERSE STOP MARKER | Rising edge while REVERSING = precise stop |
| 11 | LIDAR OUTER (NO) | ON = object · 60 % of current speed (when lidar enabled) |
| 12 | LIDAR MIDDLE (NO) | ON = object · 30 % of current speed (when lidar enabled) |
| 13 | LIDAR INNER (NO) | ON = object · stop at once (when lidar enabled) |
| 14 | PUSHER PROX TOP | Pin at top |
| 15 | PUSHER PROX BOTTOM | Pin at bottom · ends FULL DOWN, releases tag C / 1C |

## Digital outputs

| DO | Function | Meaning / use |
|---|---|---|
| 0 | LEFT driver FWD | Direction forward |
| 1 | LEFT driver REV | Direction reverse |
| 2 | LEFT driver BRAKE | Brake |
| 3 | LIDAR AREA SEL 1 | Lidar field set (set_lidar_area), OFF by default |
| 4 | RIGHT driver FWD | Direction forward |
| 5 | RIGHT driver REV | Direction reverse |
| 6 | RIGHT driver BRAKE | Brake |
| 7 | LIDAR AREA SEL 2 | Lidar field set (set_lidar_area), OFF by default |
| 8 | HORN AUTO | ON while RUNNING or REVERSING (OFF while the ALARM horn is on) |
| 9 | HORN ALARM | ON when the trolley is not caught 10 s after the CC / marker stop |
| 10 | spare | — |
| 11 | AT HOMEPOST | ON while parked at homepost |
| 12 | PUSHER DOWN a | Relay K1 · with DO15 |
| 13 | PUSHER UP a | Relay K2 · with DO14 |
| 14 | PUSHER UP b | Relay K3 · with DO13 |
| 15 | PUSHER DOWN b | Relay K4 · with DO12 |

### Analog outputs (AO module 192.168.1.30, register base 4096)

| AO | Function | Range |
|---|---|---|
| 0 | LEFT motor speed | Code sends 0–5 V (driver 0–10 V → 0–3000 rpm) |
| 1 | RIGHT motor speed | Code sends 0–5 V (driver 0–10 V → 0–3000 rpm) |

### Network, CAN and USB

| Device | Link | Address |
|---|---|---|
| IPC (Ubuntu) | enp7s0 → IO modules | 192.168.1.1 |
| IPC (Ubuntu) | enp4s0 → RFID reader | 192.168.3.100 |
| DIO module | Modbus TCP | 192.168.1.31:502 · unit 1 |
| AO module | Modbus TCP | 192.168.1.30:502 · unit 1 |
| RFID reader | TCP (built-in network module) | 192.168.3.200:2022 |
| MGS1600 front | CAN 500 kbit/s (CANable2 USB) | Node 6 · TPDO 0x186 |
| MGS1600 rear | CAN 500 kbit/s (CANable2 USB) | Node 7 · TPDO 0x187 · REVERSE only |
| Remote / keypad | 2.4 GHz USB dongle on the IPC | “XING WEI 2.4G” (profile keypad.NAME_CONTAINS) |
| Web HMI | HTTP | <ipc>:5000 (also opens full-screen on the IPC screen) |

## 6. Reverse and trolley hitch (pick-up)

| Step | Trigger | What happens |
|---|---|---|
| 1 | Tag 0001 (RUNNING) — branch 2: tag 0002 | Stop, wait 2 s, switch to REVERSE (skipped while ignore_rfid_1 / ignore_rfid_2 is set). Branch 2 uses only 0002 (starts right, AE / BB switch the track) and ignores 0001 / 0003; other branches ignore 0002. Tag 0003 (all branches except 2) forces the LEFT track. |
| 2 | — | Reverse at 0.15 m/s on the rear sensor. Track fixed by branch (`reverse_track_by_branch`): branch 1 (DI4) and 3 (DI6) = **right**; AE / BB are ignored for them. Branch 2, 4 / no branch: starts right, 00AE → left, 00BB → right. |
| 2b | Tag 00BA | Arms the DI10 stop marker. Before BA the marker is ignored (logged); disarmed at every reverse start. |
| 3 | Tag 00CA | Slow to 0.1 m/s for the approach. |
| 4 | DI10 marker (only after 00BA) or tag 00CC (first wins) | Brake and hold, still in REVERSE. Set ignore_rfid_1 / 2 / 3. |
| 5 | DI9 OFF (trolley caught by the limit switch) | Wait — no timeout. **Not caught after 10 s → horn switches from AUTO (DO8) to ALARM (DO9)** until DI9 goes OFF or RESET. |
| 6 | +2 s | Pusher full UP automatically (DO13 + DO14). |
| 7 | DI9 OFF AND DI14 ON (pusher top) | Leave REVERSE, drive forward at **EXTRA_SLOW 0.15 m/s** (until the next speed tag). No timeout (RESET to abort). |
| 8 | — | Tag 00CC ignored for 5 s after moving off; tags 0001 / 0002 / 0003 ignored until tag 00CE. |

## 7. Trolley release — tag C, or tag 1C on branch 2 (drop-off)

Branch 1, 3, 4 or no branch (e.g. a CALL START run) drop the trolley at **C (000C)**. Branch 2 drives past C and drops it at **1C (001C)**. The steps are the same:

| Step | Trigger | What happens |
|---|---|---|
| 1 | Tag 000C, or 001C on branch 2 (RUNNING) | Stop and brake. |
| 2 | — | Pusher DOWN automatically (DO12 + DO15). |
| 3 | DI15 ON (pusher bottom) | Pusher stops. No timeout — the AGV waits until DI15 is ON (RESET to abort). |

| Step | Trigger | What happens |
|---|---|---|
| 4 | +2 s | Drive forward at SLOW. |
| 5 | Tag 0000 | Stop at homepost, clear the branch, back to ARMED, DO11 ON. |

# 8. Pusher

- UP = DO13 + DO14, DOWN = DO12 + DO15, through a 4-channel relay board wired as an H-bridge. The code never energises both pairs together.
- **Pusher button DI0** (ARMED or MANUAL only): pin at bottom → full UP; at top → full DOWN; in between → opposite of its last move; press while moving → stop.
- **Keys Z / X** (ARMED or MANUAL only): full UP / full DOWN.
- Web FULL UP / FULL DOWN stop at DI14 / DI15 or after 15 s. JOG UP / DOWN ignore both proximity sensors. PUSHER E-STOP latches until cleared.
- At startup all pusher outputs are forced OFF; the pin is not moved. Trolley limit switches are in use: “hooked” = DI9 OFF and DI14 ON.

# 9. Lidar safety fields (AUTO forward only, NO outputs)

| Input | Object in field | Effect |
|---|---|---|
| DI11 outer | ON | Ease down to 60 % of the current speed |
| DI12 middle | ON | Ease down to 30 % of the current speed |
| DI13 inner | ON | Brake immediately; when clear, wait 2 s, then continue |

- **OFF by default** (`lidar.ENABLED: false`). Turn it on with PARAMS → LIDAR SAFETY (engineer login, AGV stopped); that switch resets to the profile value at restart. While off, a red **LIDAR OFF** badge shows on the HOME page.
- Percentages are `OUTER_PERCENT: 60` / `MIDDLE_PERCENT: 30` in the profile.
- Outputs are normally open (`NORMALLY_CLOSED: false`): OFF = clear, ON = object. With NO wiring a broken cable reads as clear.
- Never raises speed above the tag speed. Not active while REVERSING or in MANUAL. DO3 / DO7 select the lidar field set via `set_lidar_area` (none assigned; both OFF).

# 10. Keyboard and USB remote control

The same keys work on the **laptop keyboard** (web MANUAL page, after clicking **KEYBOARD ON**) and on the **2.4 GHz remote / keypad** whose USB dongle is plugged into the AGV PC (found automatically, hot-plug).

| Key | Action | Works in |
|---|---|---|
| W A S D · arrows · remote ring | Hold to drive: forward / spin left / reverse / spin right; two keys = curve; release = stop | MANUAL |
| 1 2 3 4 5 | AUTO speed APPROACH / SLOW / MEDIUM / HIGH / MAX | ARMED, RUNNING |
| Z / X | Pusher full UP / full DOWN | ARMED, MANUAL |
| R | AUTO REVERSE start (ARMED) / stop (REVERSING) | ARMED, REVERSING |
| Q | Switch MANUAL ↔ AUTO (overrides the DI2 selector until it is moved) | Any |
| 7 8 9 0 | Start AUTO to branch 1 / 2 / 3 / 4 (other modes: select only) | ARMED |
| T / Y | Rotate 180° on the tape LEFT (CCW) / RIGHT (CW); T or Y again = stop | MANUAL, ARMED |
| E | EMERGENCY STOP — latched until RESET | Any |
| ENTER · remote OK | RESET (same as DI1) | Any |

- Driving uses a 400 ms watchdog: the AGV stops if the key is released, the browser loses focus, the dongle is unplugged or the link drops.

- Only the remote named in keypad.NAME_CONTAINS (“XING WEI”) is used; its keys go only to the AGV (GRAB: true), not to the AGV PC desktop. python3 debugging/list_keypads.py lists connected keyboards. Turn the remote's air-mouse off.
- The service user needs group input (build_release.sh does this). Library: evdev.

# 11. Manual mode, homepost and horn

- Web jog pad, keyboard or remote: forward/reverse at 0.4 m/s, turns and spins at 0.2 m/s, smooth ramps at 0.8 m/s². E-stop still cuts power immediately.
- DO11 AT HOMEPOST: ON while parked at homepost. At power-up the AGV is assumed to be at homepost.
- **Brake when stopped** (brake.ACTIVE_WHEN_STOPPED: true): DO2 / DO6 brake ON whenever the AGV stands still, in every mode incl. EMERGENCY. Set false to release the motors (push by hand).
- DO8 horn AUTO: ON while RUNNING or REVERSING. DO9 horn ALARM: replaces DO8 when the trolley is not caught within 10 s at the CC / marker stop; RESET silences it.

## Rotate 180° on the tape (rotate.py, profile section rotate)

- Start: key **T** = left (CCW), **Y** = right (CW), or the **ROTATE 180°** buttons on the MANUAL page. Allowed in MANUAL or ARMED, with tape under the front sensor and **no trolley hooked** (BLOCK_WITH_TROLLEY: true).
- The AGV spins in place at SPEED_MPS: 0.10 (wheel speed). The front sensor (7.6 cm ahead of the axle) loses the tape at ~45°, finds it again at ~135°, then the AGV creeps at CREEP_MPS: 0.04 and brakes when the tape is within CENTER_TOL_MM: 5 of centre (≈ 180°).
- Afterwards it **stops and waits**, braked, in the mode it came from. Stop a turn with T / Y again, web STOP, RESET or E-stop. Aborts (brake) if the tape is not found before ~MAX_ANGLE_DEG: 260° or TIMEOUT_S: 40 s, or if the front sensor stops responding. Lidar INNER (when enabled) pauses the turn.
- Use it on a plain straight tape, away from forks, crossings and RFID tags. Spin / creep speed, centre tolerance and the angle window are on PARAMS → ROTATE 180°.

# 12. Web dashboard (http://<ipc>:5000)

| Page | Functions | Access |
|---|---|---|
| HOME | Mode, sensors, PID values, RFID, branch selector, sequence status, LIDAR OFF badge | Open |
| MANUAL | Jog pad, KEYBOARD panel (key list, E-stop banner, remote status), bypass buttons, pusher panel, AUTO REVERSE, ROTATE 180° | Open |
| IO | Live DI/DO lamps | Open |
| IO TEST | Force any DO ON/OFF (auto-exits when the AGV drives) | Engineer |
| PARAMS | Feature switches; live tuning (APPLY saves as default); **PID AUTO-TUNE**; **RUN LOGS** (CSV + plot downloads) | Engineer |
| Calibration | Wheel-speed calibration ramp | Engineer |
| ERRORS | Error log, controller restart button | Open |

- **Kiosk:** after kiosk/install_kiosk.sh the AGV PC logs in automatically and opens the dashboard full-screen at power-on (Alt+F4 to leave).
- Engineer login: ENGINEER button in the bottom bar, 30-minute session. Set the password with sudo AGV_ID=agv2_kim /opt/agv2/main.bin --set-engineer-password (release) or AGV_ID=agv2_kim python3 main.py --set-engineer-password (source).
- Saved tuning: /opt/agv2/data/tuning_agv2_kim.json (release) or data/ in the source folder; it overrides the profile for those values.

# 12b. PID logging and auto-tune

- **Run logs:** every AUTO run and auto-tune writes a CSV (time, speed mode, raw sensor left/right, error, P/I/D, output, speed reduction, wheel rpm, gains in use) and a 3-panel plot to analysis_plot/. Download them from PARAMS → RUN LOGS.

- **PID auto-tune** (PARAMS, ARMED, straight tape): accelerates to the chosen mode's speed, runs a short relay test (gentle weave, up to 10 s), brakes, fits a steering model (gain, drive lag, sensor look-ahead `kinematics.SENSOR_LOOKAHEAD_M = 0.076`) and proposes KP / TD / N as Calm / Recommended / Responsive. APPLY writes them like live tuning. Every speed mode (SLOW, MEDIUM, HIGH, MAX) has its own gain set; choose **SLOW BRANCH** to tune the AA / AB / AC branch gains (runs at SLOW speed — do it on the branch tape if possible).
- Aborts on tape loss, CAN timeout, lidar inner field, RESET, emergency or STOP. RFID tags are ignored during the test.

# 13. RFID reader

CP-203910 / R121MU-910 UHF reader, CF frame protocol, **Active mode** (sends tags by itself). Tags: EL-UHF-TML7007M on-metal labels (Impinj Monza R6-P).

| Setting | Value | Why |
|---|---|---|
| Work mode / interface | Active / RS232 | RS232 is the reader's internal link to its built-in network module — keep it |
| RF power | 33 dBm | Maximum |
| Q value | 1 | Fast for one tag at a time |
| Query interval | 1 (×10 ms) | Short pause between scans |
| Session / filter time | S0 / 0 | — |
| Buzzer | off | No beep on every read |

- Code: frames are decoded byte-wise with length and CRC check (old “split on CF” bug lost tags containing C/F). The reader's info reply is no longer read as a tag.
- Tools (stop the controller first): `debugging/rfid_settings.py read|set|restore`, `rfid_rate_test.py`, `rfid_sweep_test.py`.
- **Installation matters most:** keep the reader and antenna away from the battery (15–20 cm or more) and from motor/battery cables; on-metal tags need metal behind them.

# 14. Open items

**Note**
• Replace all placeholder RFID tag values (incl. new 001C and 00A2) with the real scanned hex.
• Tags are missed above ~0.2 m/s on this AGV: move the reader/antenna away from the battery, or use 2–3 copies of each tag in a row.
• TITIK 4 (00AD) has no track action yet. The branch is no longer cleared at BB: fork tags passed again on the way home act again for that branch.
• Reverse track is untested: if the AGV takes the wrong track backwards, swap left/right in reverse_track_by_branch (branches 1–3) or in reverse_fork_ae / reverse_merge_bb (branch 4).
• Lidar is NO: a broken lidar cable reads as clear. NC outputs are safer if the lidar supports them.
• Pusher relay protection not added yet (UP/DOWN interlock on IO TEST, dead-time on direction change). Fit a fuse on the relay supply.
• MAX 1.0 m/s is above the motor's ~0.94 m/s capability and leaves no steering headroom. Spare channels: DO10 only.